

Amal Mehta

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ML research engineer focused on models that adapt under distribution shift (meta-RL, sim-to-real transfer, policies deployed on physical robots), driven by a long-standing interest in how the brain learns; prior neuroimaging research at UCSF.

Experience

Independent ML & Robotics Research

Sept 2025 – Present

Self-directed projects; details at amalmehta.github.io/PortfolioWebsite

San Francisco Bay Area

- **Neural decoding:** decoded 2D hand velocity from 152 motor and premotor cortex neurons (NLB MC_Maze); GRU decoder reached R^2 0.75 on held-out trials vs. 0.55 for a Kalman filter.
- **Neuroimaging Platform:** Mac app that loads 3D brain MRI, segments gray matter, white matter and CSF voxel by voxel, and places EEG electrodes on the head model.
- **Autonomous Driving Sim:** browser self-driving stack covering perception, prediction, planning and control, with a learned speed policy gated by a safety validator; tested on 60 drives with every failure traced to its cause.
- **Sport Analyzer:** real-time 3D pose estimation from a webcam to measure running stride, lifts and swings; scores a user's movement against a single reference rep.
- **Sensorimotor AI Journal Club, Redwood Center for Theoretical Neuroscience, UC Berkeley:** member.

Deep Learning Scientist

Sept 2021 – Aug 2025

PhysicsAI (DARPA-funded)

Pacifica, CA (remote)

- Formulated synthetic-data generation as RL: policies (TD3, PPO, DDPG) set image-generator parameters, rewarded by detector performance on real drone imagery, closing the sim-to-real gap without hand-tuned rendering.
- Built a PEARL meta-RL agent (probabilistic task embeddings over TD3) for few-shot adaptation across generator task distributions.
- Implemented domain adaptation from scratch to hold detector performance in the low-real-label, high-synthetic-data regime; built GAN-based synthesis pipelines (DCGAN, InfoGAN, FineGAN) producing training data for detection models.

Deep RL Researcher

Aug 2023 – Mar 2024

Hybrid Robotics Lab, Mechanical Engineering, UC Berkeley

Berkeley, CA

- Trained a single multi-stage PPO policy in IsaacGym for a Unitree Go1 quadruped to track, approach, and shoot a soccer ball at a target, replacing the multi-policy hierarchical framework used in prior work; ~60% kicking accuracy on hardware.
- Designed the stage-transition reward machinery and CPG-based gait tracking; prototyped RLPD (SAC with prior data) to fine-tune the sim-trained policy on real robot data.

Research Assistant

June 2020 – Dec 2022

Video & Image Processing Lab, BAIR — adv. Prof. Avidesh Zakhor

UC Berkeley

- Developed a deep CNN segmentation pipeline for invasive melanoma in large-scale histopathological images; published at SPIE Medical Imaging 2023 and presented at USCAP Annual Meeting 2022 (Stowell-Orbison Award Poster Session).

Researcher

June 2020 – Aug 2020

Abbasi-Asl Lab, UCSF — adv. Prof. Reza Abbasi-Asl

San Francisco, CA

- Built empirical mode decomposition (EMD) pipelines for 2D and 3D brain image registration on volumetric neuroimaging data.

Education

University of California, Berkeley

Aug 2017 – Aug 2021

B.A. Computer Science — Regents Scholar

Berkeley, CA

- Few-shot imitation learning: RL^2 from scratch and inverse soft Q-learning in Meta-World, benchmarked vs. iMAML and BC.

Stanford University

Aug 2021 – May 2023

Graduate Certificates — Autonomous Systems & Artificial Intelligence

- CS231N (Deep Learning for Vision), CS330 (Meta-Learning), AA271A/B (Autonomy I & II), AA273 (State Estimation).

Technical Skills

Languages

Python, C++, C

ML / RL

PyTorch, TensorFlow, stable-baselines3, Gym, Meta-World, IsaacGym, ONNX, OpenCV

Robotics

ROS, Unitree Go1, CPG gait control, Kalman filtering, 3D pose estimation

Tools

Git, GitHub Actions CI, Linux, Docker, CUDA/multi-GPU, W&B, TensorBoard

Publications

1. Shah, A., **Mehta, A.**, Wang, M., Neumann, N., McCalmont, T., Zakhor, A. “Deep Learning Segmentation for Invasive Melanoma.” *SPIE Medical Imaging*, 2023.
2. Neumann, N., Wang, M., **Mehta, A.**, et al. “Quantifying Invasive Melanoma Volume by Deep Learning Segmentation at the Pixel-Level.” *Stowell-Orbison Award Poster Session, USCAP Annual Meeting*, 2022.
3. **Mehta, A.**, Asthana, S., Chowdhary, P., Cappi, J., Nakamura, T. “Contract Information Extraction Using Watson NLP and Human-in-the-Loop.” *INFORMS Annual Meeting*, 2021.